PRevIews

Searching Films by What Their Reviews Say

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The Problem

Film ratings tell us how much people liked a film, not why.

Why Search by Reviews?

- Film reviews describe plot, atmosphere, themes, and what reviewers praised/criticised
- Metadata and star ratings cannot capture this richness
- Existing sites only allow search by title, cast, genre, or score
- Gap: No system lets users search by review content



The PReviews Vision

What PReviews Does

- Find films through the content of their reviews
- Contrast two perspectives:
  - General audiences (IMDb)
  - Professional critics (Rotten Tomatoes)
- Enrich each review with film metadata
- Each film described by its English Wikipedia article

Key Features

- Searching review content rather than metadata
- Comparing critics vs. audiences through linked sources
- Recovering full text of critic reviews from critics' websites

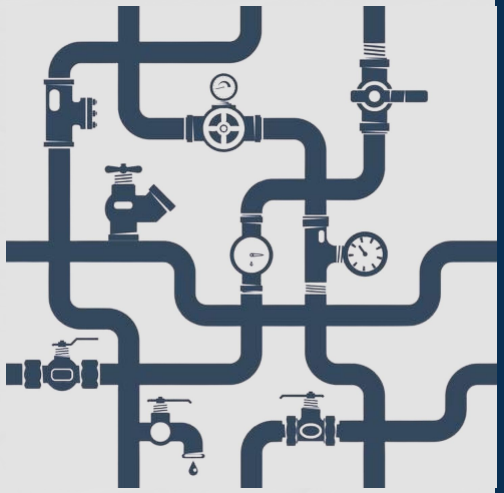
Data Sources and Integration

Source	Role	Volume
IMDb Spoiler Dataset	Audience reviews (anchor)	573,913 reviews of 1,572 films
Clapper (Rotten Tomatoes)	Critic reviews + metadata	1.44M reviews of 69,263 films
Wikidata	Links IMDb ↔ Rotten Tomatoes	90,761 pairs
English Wikipedia	Encyclopedia description	1,507 articles

Data Processing Pipeline

COLLECT → LINK → WIKIPEDIA → CLEAN → SAMPLE → MERGE → DOCUMENTS

COLLECT - Download the datasets and retrieve film identifiers.
LINK - Match films across IMDb and Rotten Tomatoes using Wikidata.
WIKIPEDIA - Download English Wikipedia articles for each film.
SCRAPE - Recover full critic reviews from their original websites.
CLEAN - Remove duplicates, fix inconsistencies, and normalize the data.
SAMPLE - Select up to 200 reviews per film and source.
MERGE - Combine the data into the final review and film documents.



Data Processing Pipeline

COLLECT → LINK → WIKIPEDIA → CLEAN → SAMPLE → MERGE → DOCUMENTS

Python · pandas · trafilatura

Python runs the pipeline, pandas processes the datasets, and trafilatura extracts text from critic webpages.

Automated with Make

Make runs all pipeline stages in the correct order with a single command.

Fixed sampling seed

A fixed seed ensures that the same reviews are selected every time the data is sampled.

Reproducible outputs

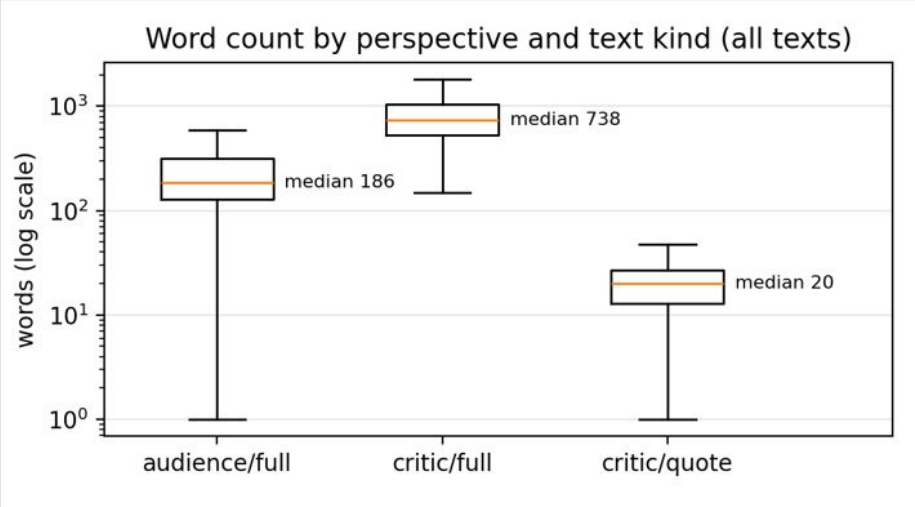
The pipeline produces the same structured files and statistics from the same input data.

Collection Scope

From short quotes to full reviews

- Rotten Tomatoes quotes: median of 21 words
- Full critic reviews: median of 738 words
- 174,305 reviews with links examined
- 56,785 full texts recovered after scraping and cleaning

Final sample: 56,637 full critic reviews

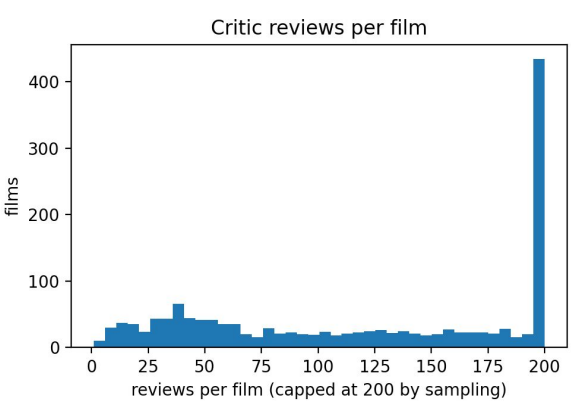


Sampling Strategy

Cap per Film	Audience	Critics	Total
50	73,529	66,880	140,409
100	142,290	115,482	257,772
200 (chosen)	259,139	179,018	438,157
No cap	548,957	207,773	756,730

Selection Method:

- Random with fixed seed (preserves opinion mix)
- Fresh share changes ≤0.25 percentage points
- Critic reviews with full text taken first



Cleaning, Sampling and Validation

Making the collection reliable

- Remove duplicates and empty texts
- Normalize dates, scores and identifiers
- Decode HTML and remove placeholders
- Limit sampling to 200 reviews per film and source
- Validate all final documents

300 randomly selected documents checked against raw data



Final Dataset and Main Results

The final collection

- 1,492 films
- 438,157 review documents
- 121.8 million words of review text
- 72% of documents contain full review text

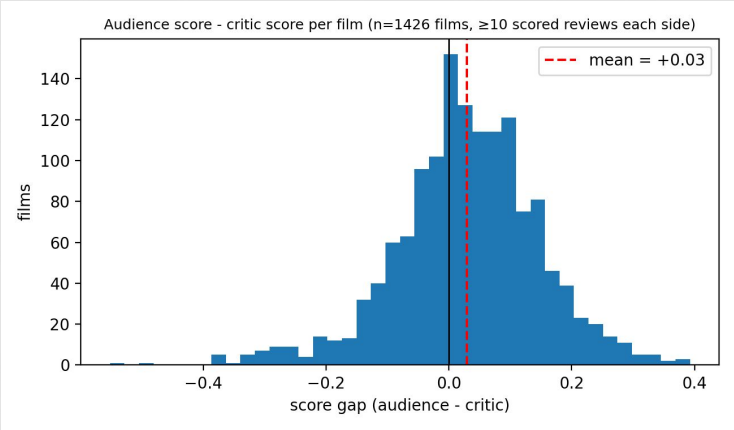
Additional findings

- 259,139 audience reviews
- 179,018 critic reviews
- 1,491 films with English Wikipedia articles

Exploratory Findings

Critics vs. Audiences:

- Average audience score: 5.1 points above Tomatometer
- 188 films rated ≥25 points higher by audiences



Kind	Documents	Median Words	Mean	Max
Audience	259,139	186	254	2,562
Critic, full text	56,637	738	942	39,264
Critic, quote	122,381	20	21	55



What Can We Search For?

From reviews to meaningful search queries

- “Films audiences love but critics dislike”
- “Atmospheric horror without jump scares”
- “Great performances in mediocre films”
- “Visually stunning films with weak scripts”

Next research questions

- Lexical vs. semantic retrieval
- Full reviews vs. short quotes
- Comparing critics and audiences
- Handling spoilers

Thanks!

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